

Modal Logic: Tableaux Examples

2026-02-18

Example 15: $((A \wedge \neg \Box A) \rightarrow$
 $\Diamond \Box (A \wedge \neg \Box A)) \rightarrow (A \rightarrow \Box A)$ in
KT

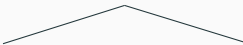
Example 15: $((A \wedge \neg \Box A) \rightarrow \Diamond \Box (A \wedge \neg \Box A)) \rightarrow (A \rightarrow \Box A)$ in KT

1. $\mathbb{F} 1, ((A \wedge \neg \Box A) \rightarrow \Diamond \Box (A \wedge \neg \Box A)) \rightarrow (A \rightarrow \Box A)$ ✓ Assumpt
2. $\mathbb{T} 1, (A \wedge \neg \Box A) \rightarrow \Diamond \Box (A \wedge \neg \Box A)$ $\rightarrow \mathbb{F} 1$
3. $\mathbb{F} 1, A \rightarrow \Box A$ $\rightarrow \mathbb{F} 1$

Assume the formula is false at world 1. The outer conditional is false, so its antecedent (line 2) is true and its consequent (line 3) is false.

1.	$\models 1, ((A \wedge \neg \Box A) \rightarrow \Diamond \Box (A \wedge \neg \Box A)) \rightarrow (A \rightarrow \Box A)$ ✓	Assumpt
2.	$\top 1, (A \wedge \neg \Box A) \rightarrow \Diamond \Box (A \wedge \neg \Box A)$	$\rightarrow \models 1$
3.	$\models 1, A \rightarrow \Box A$ ✓	$\rightarrow \models 1$
4.	$\top 1, A$	$\rightarrow \models 3$
5.	$\models 1, \Box A$ ✓	$\rightarrow \models 3$
6.	$\models 1.1, A$	$\Box \models 5$

Line 3 is false: $A \rightarrow \Box A$ is a false conditional, so A is true at 1 (line 4) and $\Box A$ is false at 1 (line 5). False $\Box$ on line 5 introduces world 1.1, where A is false (line 6).

1.	$\mathbb{F} 1, ((A \wedge \neg \Box A) \rightarrow \Diamond \Box (A \wedge \neg \Box A)) \rightarrow (A \rightarrow \Box A)$ ✓	Assumpt
2.	$\mathbb{T} 1, (A \wedge \neg \Box A) \rightarrow \Diamond \Box (A \wedge \neg \Box A)$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, A \rightarrow \Box A$ ✓	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, A$	$\rightarrow \mathbb{F} 3$
5.	$\mathbb{F} 1, \Box A$ ✓	$\rightarrow \mathbb{F} 3$
6.	$\mathbb{F} 1.1, A$	$\Box \mathbb{F} 5$
		
7.	$\mathbb{F} 1, A \wedge \neg \Box A \quad \mathbb{T} 1, \Diamond \Box (A \wedge \neg \Box A)$	$\rightarrow \mathbb{T} 2$

Line 2 is a true conditional: it branches. Either the antecedent $A \wedge \neg \Box A$ is false at 1 (left branch, line 7), or the consequent $\Diamond \Box (A \wedge \neg \Box A)$ is true at 1 (right branch, line 8).

1.	$\mathbb{F} 1, ((A \wedge \neg \Box A) \rightarrow \Diamond \Box (A \wedge \neg \Box A)) \rightarrow (A \rightarrow \Box A) \checkmark$	Assumpt
2.	$\mathbb{T} 1, (A \wedge \neg \Box A) \rightarrow \Diamond \Box (A \wedge \neg \Box A) \checkmark$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, A \rightarrow \Box A \checkmark$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, A$	$\rightarrow \mathbb{F} 3$
5.	$\mathbb{F} 1, \Box A \checkmark$	$\rightarrow \mathbb{F} 3$
6.	$\mathbb{F} 1.1, A$	$\Box \mathbb{F} 5$
7.	$ \begin{array}{cc} \mathbb{F} 1, A \wedge \neg \Box A \checkmark & \mathbb{T} 1, \Diamond \Box (A \wedge \neg \Box A) \end{array} $	$\rightarrow \mathbb{T} 2$
8.	$ \begin{array}{cc} \mathbb{F} 1, A & \mathbb{F} 1, \neg \Box A \checkmark \end{array} $	$\wedge \mathbb{F} 7$
9.	$ \begin{array}{cc} \otimes & \mathbb{T} 1, \Box A \\ & \otimes \end{array} $	$\neg \mathbb{F} 8$

1.	$\models 1, ((A \wedge \neg \Box A) \rightarrow \Diamond \Box (A \wedge \neg \Box A)) \rightarrow (A \rightarrow \Box A) \checkmark$	Assumpt
2.	$\top 1, (A \wedge \neg \Box A) \rightarrow \Diamond \Box (A \wedge \neg \Box A) \checkmark$	$\rightarrow \models 1$
3.	$\models 1, A \rightarrow \Box A \checkmark$	$\rightarrow \models 1$
4.	$\top 1, A$	$\rightarrow \models 3$
5.	$\models 1, \Box A \checkmark$	$\rightarrow \models 3$
6.	$\models 1.1, A$	$\Box \models 5$
7.	$\models 1, A \wedge \neg \Box A \checkmark$ $\top 1, \Diamond \Box (A \wedge \neg \Box A) \checkmark$	$\rightarrow \top 2$
8.	$\models 1, A$ $\models 1, \neg \Box A \checkmark$	$\wedge \models 7$
9.	$\otimes$ $\top 1, \Box A$	$\neg \models 8$
10.	$\otimes$ $\top 1.2, \Box (A \wedge \neg \Box A)$	$\Diamond \top 7'$
11.	$\top 1.2, A \wedge \neg \Box A \checkmark$	$\top \Box 8$
12.	$\top 1.2, A$	$\wedge \top 9$
13.	$\top 1.2, \neg \Box A \checkmark$	$\wedge \top 9$
14.	$\models 1.2, \Box A \checkmark$	$\neg \top 7/8$

Why is this valid?

- Assume A is true but $\Box A$ is false at w_1 : so $A \wedge \neg\Box A$ holds at w_1 , and some accessible world $w_{1.1}$ has A false.
- If the antecedent of our assumption holds – $(A \wedge \neg\Box A) \rightarrow \Diamond\Box(A \wedge \neg\Box A)$ – then $\Diamond\Box(A \wedge \neg\Box A)$ is true at w_1 .
- So some accessible world $w_{1.2}$ has $\Box(A \wedge \neg\Box A)$ true.
- By reflexivity (T condition), $A \wedge \neg\Box A$ holds at $w_{1.2}$ itself: so $\neg\Box A$ is true at $w_{1.2}$, meaning some world $w_{1.2.1}$ accessible from $w_{1.2}$ has A false.
- But $\Box(A \wedge \neg\Box A)$ is true at $w_{1.2}$ (reusable), so it fires at $w_{1.2.1}$ too: A must be true at $w_{1.2.1}$. Contradiction.

The antecedent $((A \wedge \neg\Box A) \rightarrow \Diamond\Box(A \wedge \neg\Box A))$ is a strong assumption: it says that whenever A holds but is not necessary, there is an accessible world where that same condition is boxed – guaranteed to persist everywhere accessible. But that boxed condition undermines itself.